Introduction to Demographic Methods

(7-8: Future; Digital e Computational Demography; Conclusions)

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Introduction to Demographic Methods

1 Introduction. Why study Demography? . . . Git, version control, course materials GitHub
2 Mortality
3 Fertility
4 Population projections
5 Migration
~~6 Kinship~~
7 The future, and Digital and Computational Demography
8 Wrap up + Description of examination
9 Final examination: 3 min presentation to share your selected topic + 2 min Q&A; 6 weeks to prepare a short essay and empirical analysis

Discussing previous week's materials

1) Slides, Readings

1. Any questions on my slides?
2. Let's discuss the "key reading material".
3. Example: Discuss at least 1-2 points, e.g., 1) What was surprising and counterintuitive for you? 2) What did you find that was already expected? 3) What could be extended in this reading/work?
4. Any thoughts about the additional reading materials?

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Universität Rostock

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2) Hands-on

1. Any problems in installing the requirements?
2. Any issues in running the codes in hands-on materials to reproduce the examples?
3. Did anyone extend the examples? How? What did you do and find?

Future of demographic research and Digital and Computational Demography (which already has 10+ years of history)

Future: Digital and Computational Demography

1. Thick vs. big data

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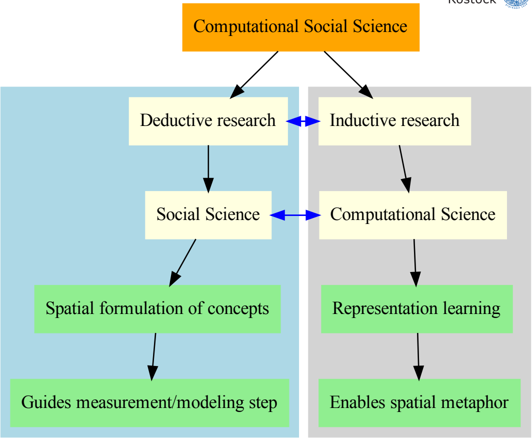
3. Inductive vs. deductive approaches to research (see diagram on the right, and Akbaritabar (2024))



See more: Akbaritabar, 2024

Future: Digital and Computational Demography

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3. Inductive vs. deductive approaches to research (see diagram on the right, and Akbaritabar (2024))
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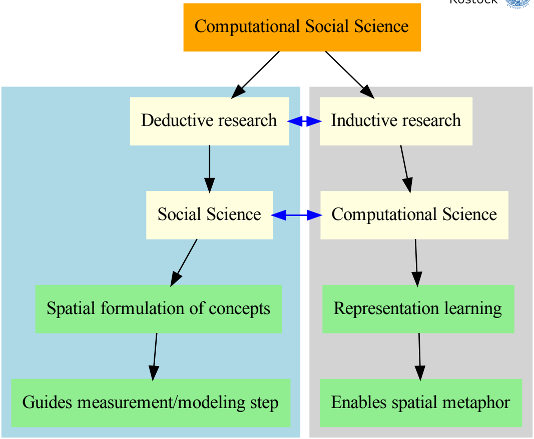

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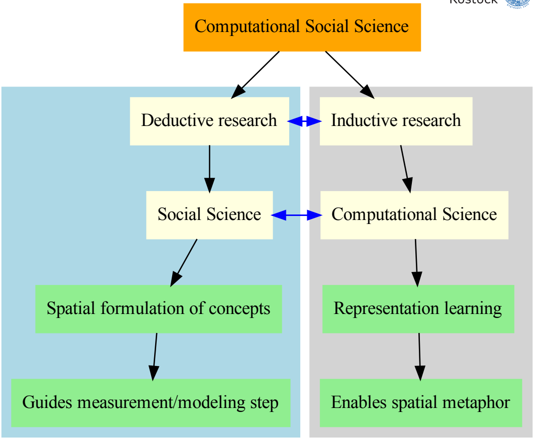
5. Mixed Methods



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Final exam

Please send me your slides for the 3 minutes presentation, at the latest, by 20:00 on the day before the session!

Where to learn basics of R and Python?

To start with Python

Check this website first:
https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:
https://github.com/vtraag/intro-python

Or this one by Data Carpentry:
https://datacarpentry.org/python-ecology-lesson/

To start with R

Check this website first:
https://www.w3schools.com/r/default.asp

This "very short introduction to R" is a good start:
https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf

Or this course by Data Carpentry:
https://datacarpentry.org/R-genomics/index.html

Reading materials for today

Key reading material

Kashyap, R., Rinderknecht, R. G., Akbaritabar, A., Alburez-Gutierrez, D., Gil-Clavel, S., Grow, A., Kim, J., Leasure, D. R., Lohmann, S., Negraia, D. V., Perrotta, D., Rampazzo, F., Tsai, C.-J., Verhagen, M. D., Zagheni, E., & Zhao, X. (2023, March). Digital and computational demography. In Research Handbook on Digital Sociology (pp. 48–86). Edward Elgar Publishing.

Additional reading materials

Salganik, M. J. (2018). Bit by bit: Social research in the digital age. Princeton University Press.

Kosinski, M., Matz, S. C., Gosling, S. D., Popov, V., & Stillwell, D. (2015).Facebook as a research tool for the social sciences: Opportunities, challenges, ethical considerations, and practical guidelines.. American Psychologist, 70(6), 543–556. https://doi.org/10.1037/a0039210

Thanks for your attention!

Questions and comments are always welcome!

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References

Akbaritabar, A. (2024). Thinking spatially in computational social science. EPJ Data Science, 13(1), 1–12. https://doi.org/10.1140/epjds/s13688-023-00443-0

Castro Torres, A. F., & Akbaritabar, A. (2024). The use of linear models in quantitative research. Quantitative Science Studies, 1–32. https://doi.org/10.1162/qss_a_00294

Torres, A. F. C., & Akbaritabar, A. (2025). The hegemonic use of linear models in quantitative social sciences. Bulletin of Sociological Methodology/Bulletin de Méthodologie Sociologique, 1–18. https://doi.org/10.1177/07591063251349381